

PART C

Practical sports medicine

The younger athlete

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Youth is full of sport, age's breath is short; Youth is nimble, age is lame; Youth is hot and bold, age is weak and cold; Youth is wild, and age is tame.

William Shakespeare (1564–1616) 'Crabbéd Age and Youth'

Participation in youth (i.e. child and adolescent) sports is popular and widespread. Many youngsters train year-round and specialise in their sports in childhood, and during adolescence some may reach peak performance and compete nationally or internationally. It is not uncommon for preteens to train 20 or more hours each week at regional training centres in tennis or gymnastics, to compete in triathlons, or for youngsters as young as 6–8 years of age to play organised hockey or soccer and travel with select teams to compete against other teams of similar calibre. Thus, knowledge about specific physiologic characteristics, management of orthopaedic conditions, injury prevention guidelines, and information regarding non-orthopaedic concerns is imperative to all individuals involved in the healthcare of young athletes.

THE YOUNG ATHLETE IS UNIQUE

Engaging in sports activities at a young age has numerous health benefits, but also involves risk of injury. Indeed, the young athlete may be particularly vulnerable to sport injury due to the physical and physiological processes of growth. Injury risk factors that are unique to the young athlete include specifics of the musculoskeletal system such as an immature skeleton, nonlinearity of growth, maturity-associated variation, the adolescent growth spurt and the response to skeletal injury.¹ Young athletes might also be at increased risk because of immature or underdeveloped coordination, skills and perception, in addition to psychological issues. Although problems do not ordinarily arise at normal levels of activity, the more frequent and intense training and competition of young

athletes today may create conditions under which this susceptibility exerts itself.

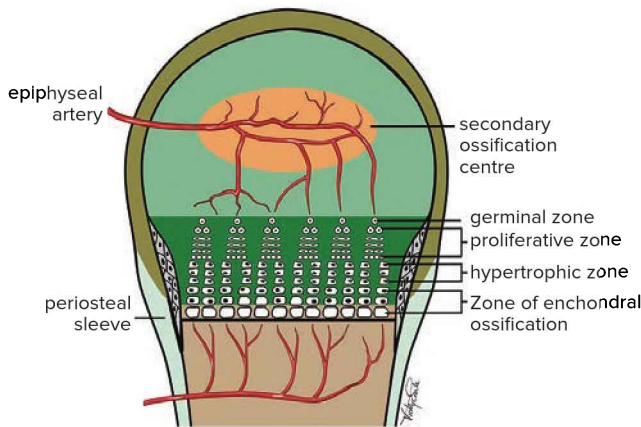
Nonlinearity of growth

The normal growth pattern is nonlinear; that is, differential growth of the body segments (head, trunk and lower extremities) occurs throughout growth and influences body proportions accordingly.² At birth, the relative contribution of head and trunk to total stature is highest and declines through childhood into adolescence. Thus, the child is characterised by a proportionally larger head and trunk, and shorter legs compared to an adult. In some events, for example rodeo 'mutton busting', one can anticipate that a young 'top-heavy' child would be at increased risk of falling off a sheep compared to an older child with proportionally longer legs.

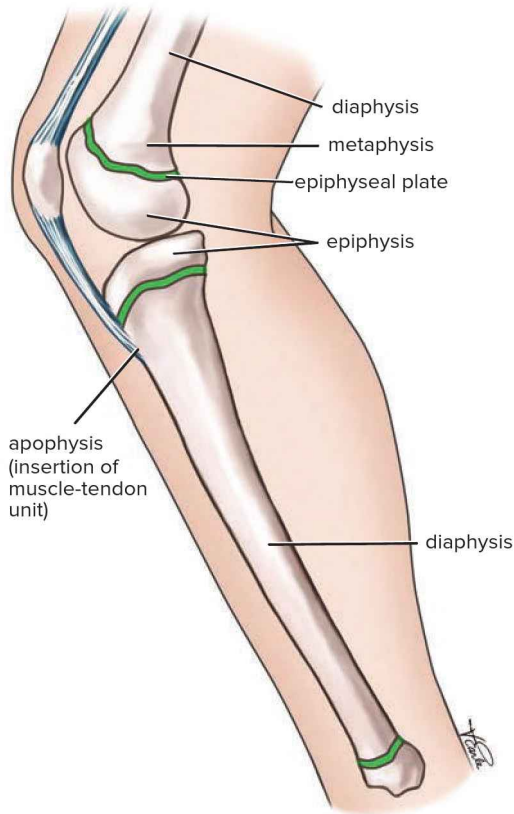
One of the features of the child's skeleton is the existence of the epiphyseal growth zones (physis). It has traditionally been divided into four zones: the resting or germinal zone, the proliferation zone, the zone of hypertrophy and the zone of enchondral ossification, which is continuous with the metaphysis (Fig. 44.1).

In each segment of the young skeleton, for example the femur or humerus, the contribution to growth of distal and proximal parts are not equal. This is the concept of the 'fertile physis'. The proximal part of the humerus and distal part of the femur contribute much more in skeletal growth, an important factor in the treatment approach after skeletal injuries.

It could also be argued that under a given physical load, for example over a distance run, a child's locomotor



(a)



(b)

Figure 44.1 (a) Anatomy of the physis (b) Different parts of the growing bone—the metaphysis, epiphysis, diaphysis and epiphyseal plate

apparatus would be exposed to greater stress: hence to a higher risk of overuse injury than that of an adult. Yet, often child athletes progress rapidly to training regimens, skills and stunts that were originally introduced and intended for more mature individuals.

Maturity-associated variation

Maturity-associated variation depends on the sex of the young athlete. In general, girls are more mature than boys at a given age. Early-maturing boys have structural, functional and performance advantages in sports requiring size, strength and power over late maturers.

PRACTICE PEARL

Children of the same chronological age may vary considerably in biological maturity status, and individual differences in maturity status influence measures of growth and performance during childhood and adolescence.²

Chronological age (CA) may add yet another dimension of individual variation given that most youth sports are categorised by CA. Within a single age group (e.g. 12 years of age), the child who is 12.9 years of age is likely to be taller, heavier and stronger than the child who is 12.0 years of age, even though both are classified as 12 years of age. The fear is that an unbalanced competition between early and late-maturing and/or older and younger boys in contact sports such as martial arts and wrestling contributes to at least some of the serious injuries in these sports. A matching system based on physical attributes, although logistically difficult to implement, may be beneficial to equalising competition, to maintaining interest in participation, and to reducing potential for injury.³

Unique response to skeletal injury

Often, the mechanism of injury in young athletes and adult athletes is the same. However, there are some significant differences in the type of injuries sustained by children and adolescents because of the differences in the structure of growing bone compared with adult bone. The different components of growing bone are indicated in Figure 44.1b.

The differences between growing and adult bone (Fig. 44.2) are summarised below.

- The periosteum is thicker in children and has considerable healing potential.
- The articular cartilage of growing bone is a thicker layer than in adult bone and can remodel.

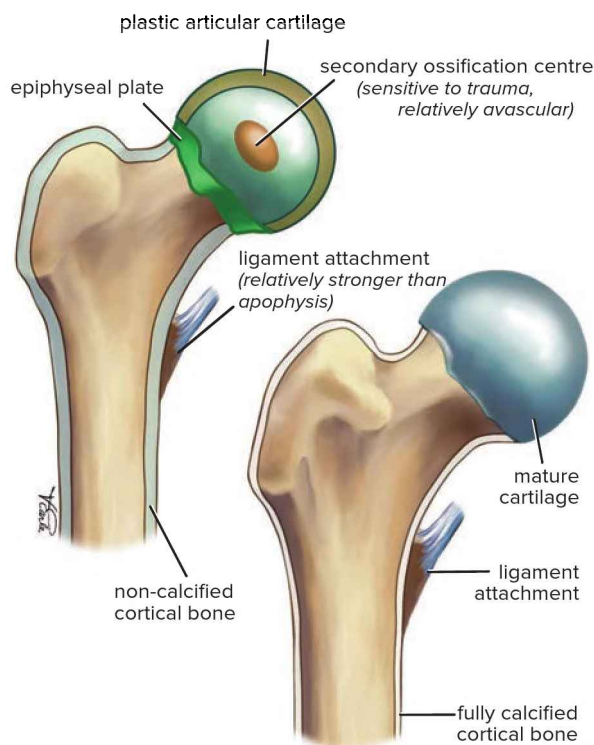


Figure 44.2 Contrasting features of growing bone (left) and adult bone (right)

- The junction between the epiphyseal plate and the metaphysis in children is vulnerable to disruption, especially from shearing forces.
- Tendon attachment sites—apophyses—are cartilaginous plates that, in children, provide a relatively weak cartilaginous attachment, predisposing to the development of avulsion injuries.
- The cortex of long bones in children is more resilient and elastic than adult bone, withstanding greater deflection without fracture. Thus, children tend to suffer incomplete fractures of the greenstick type, which do not occur in adults.
- During the adolescent growth spurt, structural changes in physeal cartilage occur that result in a thicker and more fragile epiphyseal plate.⁴ Studies of the incidence of physeal fractures indicate an increased occurrence of fractures during pubescence.^{5,6}
- It has been proposed that the adolescent growth spurt may also increase susceptibility to physeal injury by causing an increase in muscle-tendon tightness about the joints and loss of flexibility.⁷ However, this concept is controversial.⁸
- The vascular supply is different especially in very young athletes.

As a result of these differences, a particular mechanism of injury may result in different pathological conditions in the younger athlete compared with the mature adult. The younger athlete is more likely to injure cartilage and bone or completely avulse an apophysis than to have a significant ligament sprain or tendinopathy. Some examples of different injuries in children and adults that are the result of similar mechanisms are shown in Table 44.1.

PRACTICE PEARL

Younger athletes have a greater healing potential than adult athletes.

MANAGEMENT OF MUSCULOSKELETAL CONDITIONS

In children, traumatic injuries may result in fractures of the long bones or the growth plates, or soft tissue damage. Strong, incoordinate muscle contractions are more likely to lead to an avulsion fracture at the site of attachment of the muscle or tendon rather than to a tear of the muscle or tendon itself.

The osteochondroses are a group of conditions affecting secondary ossification centres of the growing bones. Although the aetiology of the osteochondroses is not well understood, non-articular osteochondroses may well be related to overuse (Table 44.2).

Each of the following common paediatric injury presentations are discussed in this chapter: acute fractures, shoulder pain, elbow pain, wrist pain, back pain or postural abnormality, hip pain, knee pain, painless abnormalities of gait and foot pain.

Acute fractures

Fractures occur in the young athlete due to the line of weakness between the epiphyseal plate and the formed bone, the relative weakness of apophyseal cartilage compared with the musculotendinous complex, and the direction and strength of the force. Four types of fractures are seen in the younger athlete: metaphyseal, physeal, diaphyseal and apophyseal avulsion fractures.

Metaphyseal fractures

Metaphyseal fractures occur especially in the forearm and lower leg. The most common type of fracture seen is a buckling of one side of the bone. The management of metaphyseal fractures in children depends on age, gender, the degree and plane of displacement, and the distance from the joint. The treatment usually simply consists of cast immobilisation and complications are very rare. Occasionally, angular or rotational deformity is present and requires open reduction and internal fixation.

Table 44.1 Comparison of injuries that occur with similar mechanisms in children and adults

Site	Mechanism	Injury in adult	Injury in child
Thumb	Valgus force as in 'skier's thumb'	Sprain of ulnar collateral ligament	Fracture of proximal phalangeal physis (usually Salter–Harris type III)
Distal interphalangeal joint of finger	Hyperflexion injury	Mallet finger	Fracture of distal phalangeal epiphysis Salter–Harris (type II or III)
Hand	Punching injury as in boxing	Fracture of metacarpal head	Fracture of metacarpal epiphysis Salter–Harris (type II)
Shoulder	Fall on point of shoulder	Acromioclavicular joint sprain	Fracture of middle third of clavicle epiphysis
	Abduction and external rotation force	Dislocated shoulder	Fracture of proximal humeral epiphysis Salter–Harris (type I or II)
Thigh/hip	Acute flexor muscle strain or extensor strain	Quadriceps strain or hamstring strain	Apophyseal avulsion of anterior inferior iliac spine or ischial tuberosity
Knee	Overuse injury	Patellar tendinopathy	Osgood–Schlatter disease or Sinding–Larsen–Johansson disease
	Acute trauma (e.g. skiing) injury	Meniscal or ligament injury	Fractured distal femoral or proximal tibial epiphysis, avulsion of tibial spine
Ankle	Acute inversion	Lateral ligament sprain	Salter–Harris type I or II fracture of tibia or fibula
Heel	Overuse	Achilles tendinopathy	Sever's disease

Table 44.2 Types of osteochondroses

Type	Condition	Site
Articular	Perthes' disease	Femoral head
	Kienböck's disease	Lunate
	Köhler's disease	Navicular
	Freiberg's disease	Second metatarsal
	Osteochondritis dissecans	Medial femoral condyle, capitellum, talar dome
Non-articular	Osgood–Schlatter disease	Tibial tubercle
	Sinding–Larsen–Johansson disease	Inferior pole of patella
	Sever's disease	Calcaneus
Physeal	Scheuermann's disease	Thoracic spine
	Blount's disease	Proximal tibia

Physeal (growth plate) fractures

Approximately 15% of all fractures in children involve the physis⁹ or growth plate, and more than a third of these occur in organised sport settings. Fractures of the growth plate are of particular concern because of the dangers of interruption to the growth process via injury to the cells in the zone of hypertrophy.

Although more elaborate systems for describing acute physeal injuries are available,⁹ the system most widely used was developed by Salter and Harris¹⁰ (Fig. 44.3). Salter–Harris types I and II fractures usually heal well. However, these injuries are not as innocuous as originally described and may occasionally be associated with local growth plate closure and osseous bridging.⁹

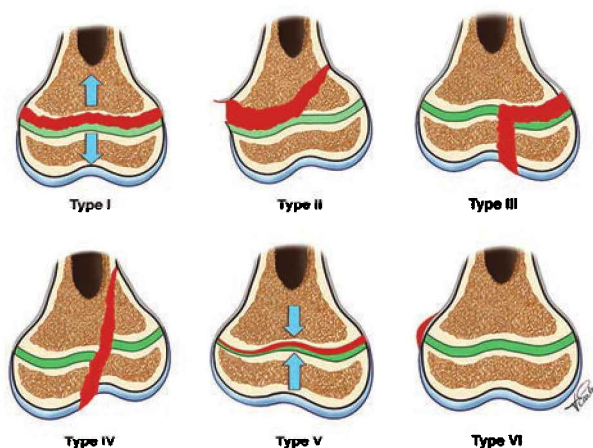


Figure 44.3 Salter–Harris classification of physal (growth plate) fractures

Type III and IV fractures involve the joint surface as well as the growth plate and have a high complication rate. If these fractures are not recognised, they could produce permanent injury to the growth plate, resulting in growth disturbances. When recognised, accurate anatomical reduction must be performed to reduce the possibility of interference in growth and to minimise the possibility of long-term degenerative change. However, occasionally the initial insult can produce permanent growth arrest despite subsequent anatomical reduction.

Type V fractures are compressive force fractures of the physis and can lead to growth zone lesions. Type VI fractures are lesions of the periosteal attachment (Ranvier zone) and can also lead to growth zone disturbances.

Management of physal fractures depends on the age and gender of the athlete, grade and plane of the displacement and the distance from the joint. In the paediatric population, they mostly require cast immobilisation only, or closed reduction and cast immobilisation (types I and II). Types III and IV may require percutaneous fixation using Kirschner wires.

PRACTICE PEARL

The common sites of physal fractures in the younger athlete with recommended management and potential complications are shown in Table 44.3.

It is most important to recognise a physal fracture. Radiographs should be obtained of both limbs (for comparison) if clinical features suggest growth plate injury. A normal radiograph does not exclude a growth plate fracture. A history of severe rotational or shear force with accompanying localised swelling, bony tenderness

and loss of function should be regarded as a growth plate fracture until proven otherwise. If there is any doubt regarding the diagnosis or management of these injuries, specialist orthopaedic referral is mandatory.

Diaphyseal fractures

Most fractures of the shaft of long bones do not involve the growth plate and metaphysis of the bone, but only the diaphysis of the long bone (especially in the forearm and lower leg). In younger athletes, fractures are often incomplete due to the thicker periosteum. They are referred to as 'greenstick' fractures. The treatment of diaphyseal fractures is usually through cast immobilisation only. Indications for surgical treatment depend on the degree of displacement, age and gender of the athlete.

Apophyseal (avulsion) fractures

Avulsion fractures occur at the apophyseal attachments of large musculotendinous units, that is, large tendons to bones or, less frequently, at ligament insertions. The common sites are at the attachment of the:

- sartorius muscle to the anterior superior iliac spine
- rectus femoris muscle to the anterior inferior iliac spine
- hamstring muscles to the ischial tuberosity
- iliopsoas tendon to the lesser trochanter of the femur
- gluteus medius to the greater trochanter of the femur
- anterior cruciate ligament to the tibial plateau
- patellar tendon to the tibial tubercle
- peroneus brevis to the base of the 5th metatarsal bone
- triceps to the olecranon
- forearm flexors and pronator to the medial epicondyle humerus.

In the younger athlete, apophyseal fractures are the equivalent of acute muscle strains in the adult. Instead of a tear of the muscle fibres in the mid-substance of the muscle or at the musculotendinous junction, the tendon is pulled away with its apophyseal attachment. Many patients will describe a 'pop' with the onset of discomfort. This is confirmed on plain radiographs (e.g. Fig. 30.17). Even though injury involving the apophyseal growth plate does not normally result in length discrepancy, angular deformity or altered joint mechanics, it may adversely affect training and performance as well as the long-term health of young athletes. The most common site of an avulsion fracture is the tibial tubercle at the attachment of the patella tendon, especially in football and basketball players. Radiographs should be performed in all cases.

Management depends on the degree of the displacement of the avulsed fragment of bone. It involves cast immobilisation and, in more displaced fragments, surgical reattachment. In general, apophyseal fractures around

Table 44.3 Management and possible complications of physeal (growth plate) fractures in young athletes

Site	Management	Potential complications
Distal radius fracture	Cast immobilisation (3–4 weeks)	Not recognised, growth disturbance
Supracondylar fracture of the elbow (undisplaced)	Sling (3 weeks)	Vascular compromise of brachial artery, median nerve damage, malalignment
Distal fibular fracture	Cast, non weight-bearing (4–6 weeks)	Growth disturbance can occur up to 18 months later
Distal tibial fracture	Cast, non weight-bearing (4–6 weeks)	Premature closure of physis can lead to angulation and leg length discrepancy
Distal femur fracture	Anatomical reduction	Greater incidence of growth discrepancies than in other fractures, Salter–Harris types I and II fractures; must be observed closely

the pelvis do not need surgical treatment because the displacement is mostly not significant and the continuity of the periosteum is preserved.

One common site of an avulsion fracture at a ligamentous attachment is at the attachment of the anterior cruciate ligament to the tibia. An acute rotational injury to the knee may present with the symptoms and signs of an anterior cruciate ligament tear. Instead of the in-substance tear common in adults, the more common injury in children is avulsion of the tibial spine or distal femoral attachment. Radiographs should be performed in all cases of acute knee injuries accompanied by haemarthrosis. Management involves surgical reattachment of the avulsed fragment and ligament.

The most commonly reported acute apophyseal sport injuries in the upper extremity involve the olecranon and medial epicondyle of the humerus. The vertebral ring apophysis is the site most often involved in the spine.

Management of avulsion fractures is identical to that for grade III tears of the muscle. It involves initial reduction of the pain and swelling, restoration of full range of motion with passive stretching and active range of motion exercises as symptoms settle, as well as a graduated program of muscle strengthening. Any biomechanical abnormalities that may have predisposed the athlete to this injury should be corrected. Reattachment is occasionally required when there is a large displacement of the avulsed fragment.

Overuse injuries of the physis

Trends in children's and youth sports include increased numbers of participants, greater duration and intensity of training, and earlier specialisation and year-round training. Increased involvement and repetition of skills practiced at an early age and continued through the years of growth raises concern about risk of injury. In particular, there is concern that the tolerance limits of the physis (growth plate) may be exceeded by the repetitive physical loading required in

many sports, particularly during the adolescent growth spurt. Physeal growth disturbance as a result of injury can result in length discrepancy, angular deformity, or altered joint mechanics, and may cause significant long-term disability.

Physeal stress injuries are thought to develop when repetitive loading of the extremity disrupts metaphyseal perfusion which may inhibit ossification of the chondrocytes in the zone of provisional calcification. The hypertrophic zone continues to widen as the chondrocytes transition from the germinal layer to the proliferative zone. Widening of the physis may be seen radiographically, whereas physeal cartilage extension into the metaphysis has been shown with magnetic resonance imaging (MRI).

Physeal stress injuries were first recognised in baseball pitchers and are also found in other young athletes involved in overhead sports, including cricket, gymnastics, badminton, swimming and volleyball.

Treatment for physeal stress injury is straightforward: rest from loading of the extremity. However, in cases involving growth disturbance, corrective surgery may be required. Clinicians need to educate parents and coaches as to the existence of overuse physeal injury and the need for rest to ensure proper recovery and return to sport participation.

Shoulder pain

Acute trauma to the shoulder in children may result in fracture of the proximal humerus, the clavicle or proximal humeral physis. Fractures of the acromion or the coracoid process are rare.

Dislocation of the glenohumeral joint is common in the adolescent but uncommon in the younger child. Dislocations in adolescents are associated with a high incidence of recurrence and development of post-traumatic instability. The management of acute dislocation of the shoulder is discussed in Chapter 24.

Stress fracture of the proximal humeral physis (growth plate) was first observed in young baseball pitchers, but

has since been reported in young athletes representing a variety of sports including cricket, gymnastics, badminton, swimming and volleyball.¹¹ Radiographs may show widening of the proximal humeral physis. Metaphyseal sclerosis and demineralisation or fragmentation of the epiphysis may also be present.¹² MRI should be performed if the X-ray is negative and clinical suspicion high. In most cases, subjects improved with rest and were able to return to their sport.

Shoulder impingement is also seen in the younger athlete. In the young athlete involved in throwing sports, the impingement is usually secondary to atraumatic instability, which develops because of repetitive stress to the anterior capsule of the shoulder joint at the end range of movement (Chapter 24). Impingement and rotator cuff tendinopathy also occur in swimmers where excessive internal rotation causes a tendency to impinge. The aetiology of these problems in throwers and swimmers is discussed in Chapters 8 and 9.

Elbow pain

Delineating injury patterns to the elbow in children can be challenging, given the cartilaginous composition of the distal humerus and the multiple secondary ossification centres which appear and unite with the epiphysis at defined ages. The pitching motion in baseball, serving in tennis, spiking in volleyball, passing in American football, and launching in javelin throwing can all produce elbow pathology caused by forceful valgus stress, with medial stretching, lateral compression and posterior impingement.

The valgus forces generated during the acceleration phase of throwing (Chapter 8) result in traction on the medial elbow structures and compression to the lateral side of the joint. This may injure a number of structures on the medial aspect of the joint. Injuries include chronic apophysitis of the medial epicondyle, acute or chronic strain of the ulnar collateral ligament (also referred to as the medial collateral ligament) or avulsion fracture of the medial epicondylar apophysis. The ulnar nerve may also be damaged.

PRACTICE PEARL

Acute ulnar collateral ligament (UCL) injuries are increasing in young baseball pitchers at an alarming rate. Children who are still growing are more likely to injure the growth area of the elbow, but older adolescents are more likely to present with a frank UCL tear.

Some young athletes have avulsion injuries where the ligament stays intact but pulls off a piece of bone. If the UCL is torn, non-operative treatment is attempted but if the ligament fails to heal, the patient often undergoes reconstructive Tommy John surgery if he or she wishes to continue throwing at a high level (Chapter 25).¹³

The lateral compressive forces may damage the articular cartilage of the capitellum or radial head. The long-term sequelae of these repetitive valgus forces include bony thickening, loose body formation and contractures.

Flexion contractures can occur because of repeated hyperextension. The majority of these contractures are relatively minor (less than 15°). Significant contractures (greater than 30°) should be treated with active and active-assisted range of motion exercises accompanied by a lengthy period of rest (e.g. 3 months).

Osteochondritis dissecans of the capitellum is also seen in pitchers and, more commonly, in gymnasts. Osteochondritis dissecans is a localised area of avascular necrosis on the anterolateral aspect of the capitellum. Initially, the articular surface softens and this may be followed by subchondral collapse and formation of loose bodies in the elbow. X-rays will frequently demonstrate the lesion. In cases with suspicious radiographic changes, MRI is critical to confirm the diagnosis as well as to characterise the extent and stability of the osteochondral lesion. The early stages of osteochondritis dissecans may respond well to rest. Surgical procedures include removal of loose bodies, joint debridement and fixation of stable fragments. The results of surgical management of this condition are variable.¹⁴

The younger child (under 11 years) may develop Panner's disease. This self-limiting condition is characterised by fragmentation of the entire ossific centre of the capitellum. Loose bodies are not seen in Panner's disease and surgery is not required.

Wrist pain

Wrist pain in children may be acute or chronic. In acute conditions, it is mostly related to a distal radial fracture/epiphyseolysis or, rarely, scaphoid fracture.

Chronic dorsal wrist pain is commonly seen in gymnasts where pain is aggravated by weight-bearing with the wrist extended. The gymnast complains of tenderness over the dorsum of the hand and perhaps swelling. An often reported cause among gymnasts is a stress injury to the distal radial physis involving compromise of the metaphyseal and/or the epiphyseal blood supply.¹¹ Longstanding injuries are associated with typical radiographic changes, including widening, irregularity, haziness or cystic changes of the growth plate. Other causes of dorsal wrist pain include scaphoid impaction syndrome, dorsal impingement syndrome/capsulitis, tear of the triangular fibrocartilage complex and stress fractures.

Management of the younger gymnast with dorsal wrist pain includes relative rest, and splinting. Strengthening of the wrist flexors may also be useful in association with tape and pads to decrease hyperextension of the wrist. Most gymnasts with stress injuries involving the distal radial physis recover with rest; however, there are also

reports of stress-related premature closure of the distal radial growth plate leading to positive ulnar variance.¹⁵

Kienböck's disease of the wrist (Chapter 26) is an osteochondrosis of the lunate bone. It occurs generally in older patients (20 years old) and rarely in adolescents.

Back pain and postural abnormalities

Younger athletes may present with pain or postural abnormalities, such as 'curvature' of the spine, or both.

Low back pain

Common causes of back pain in the younger athlete are similar to those in the mature adult. Minor soft tissue injuries to the intervertebral disc, the apophyseal joints and associated ligaments, and muscle strains in the paravertebral muscles usually respond well to reduction in activity. Manipulative treatment in the management of these conditions in the younger athlete is probably contraindicated.

Back pain is more common among young athletes participating in sports with high demands on the back, such as wrestling, gymnastics and American football, than other athletes or non-athletes.¹⁶ Back pain in the young athlete may be associated with one or more of the following conditions: idiopathic scoliosis, spondylolysis, spondylolisthesis, disc degeneration, disc herniation, Schmorl's nodes, vertebral end-plate fracture and atypical Scheuermann's lesion (vertebral apophysitis). The spine, as with the rest of the skeleton, is at greater risk of injury during the adolescent growth spurt.¹⁶

Severe disc injuries and tumours are occasionally seen in the lumbar spine of the adolescent athlete. Biomechanical abnormalities such as leg length discrepancy, pelvic instability and excessive subtalar pronation may also indirectly lead to low back pain.

Idiopathic scoliosis

Idiopathic scoliosis accounts for 80% of structural scoliosis and is the most common cause of back discomfort in young athletes. Nearly 32% of adolescents with idiopathic scoliosis complain of back discomfort.¹⁷ The pain is not related to the size or location of the curve and usually does not interfere with the patient's ability to perform daily activities.

Non-predictive but predisposing factors for the progression of the scoliosis before skeletal maturity include a family history of scoliosis, patient height-to-weight ratios, lumbosacral transitional anomalies, thoracic kyphosis, lumbar lordosis and spinal balance.

In general, the rate of progression of scoliosis in adulthood is much slower than that in adolescence and depends on the size of the curve once skeletal maturity has been reached.

The screening protocol for scoliosis in young athletes is the most important issue in the assessment of spine disorders. On physical examination the most common findings are: shoulder asymmetry, unequal scapular

prominence, appearance of an elevated or prominent hip, greater space between the arm and body on one side (with the arms hanging loosely on the side), head not centred over the pelvis, and a positive forward-bending test (Adam's test). A plain radiograph in anteroposterior and lateral view plays an important part in assessment.

In selecting appropriate treatment, the clinician must take into consideration the adolescent's remaining growth potential, the severity of the curve, as well as the pattern and location of the scoliosis. The available treatment choices are observation, non-surgical intervention and surgical intervention. It is imperative that clinicians know which option is appropriate for each individual patient.

Actively growing adolescents (Risser grade 2 or lower) with curves between 30° and 45° should be started on brace therapy at the time of the initial visit. In very immature patients (Risser grade 0 and pre-menarcheal if female) with curves exceeding 25° degrees, bracing should be started immediately. In most cases, growing adolescents with curves exceeding 45-50° require operative treatment (stabilisation) because other forms of treatment are ineffective to control or correct the scoliosis. Skeletally mature individuals with curves exceeding 50-55° also are at risk for curve progression and should be considered for surgical treatment.

Spondylolysis

Spondylolysis has received particular attention due to its high prevalence among athletes (Chapter 29). This condition may also occur in the younger athlete, particularly because of repeated hyperextension of the lumbar spine. This injury is typically seen as a result of participation in ballet, gymnastics, diving, volleyball, fast bowling in cricket and serving in tennis. The management of this condition is discussed in Chapter 29. The amount of hyperextension activity must be reduced and this may involve some alteration in technique.

There is considerable debate whether these defects in the pars interarticularis are congenital or acquired. They are probably acquired, even though this may occur at an extremely young age. A fibrous union develops across the defect and this is susceptible to injury. The presence of a pars interarticularis defect does not automatically mean that this is the cause of the patient's pain. MRI may confirm that the pars interarticularis defect is the site of an acute fracture.

Spondylolisthesis

Spondylolisthesis is defined as the forward slippage of one vertebra on its adjacent caudal segment (Chapter 29). Spondylolisthesis is one of the most variable conditions affecting the paediatric spine. The severity of its clinical manifestation ranges from a coincidental radiographic finding in asymptomatic patients, to a disabling deformity that produces severe postural and gait disturbances, pain

and neurologic impairment affecting the lower extremities, the bowel and the bladder. Just as the initial symptoms can vary, so does the radiographic severity of the slippage, which ranges from a few millimetres of anterior displacement to complete dislocation of L5 anteriorly over the sacrum (called *spondyloptosis*). Unabated controversy continues regarding optimal surgical treatment in severe cases.

Spondylolisthesis is divided into five types based on the radiographic findings and age at onset: the dysplastic, isthmic, degenerative, post-traumatic and pathologic types. The age of the patient and the degree of listhesis determine whether observation, non-operative management, or immediate surgical stabilisation is necessary. It has been commonly believed that the younger the patient, the greater the risk of progression of the deformity.

Low-grade spondylolisthesis (<50% slip) is treated conservatively, with surgical treatment reserved for patients with continued symptoms, disability, or in the case of juvenile onset, evidence of progressive listhesis. High-grade spondylolisthesis (>50% slip) generally requires surgical treatment. Objective opinions that reduction of the slip angle and L5 incidence correlate with better clinical outcome suggest that this is becoming the treatment of choice, although the degree of reduction is still being debated. Current expert opinion favours *partial* reduction—by postural means if possible—maintained by internal fixation as the appropriate middle ground to achieve neurologic safety, sagittal realignment and a high fusion rate.¹⁷

Postural abnormalities

The commonest postural abnormality of the spine in the younger athlete is excessive kyphosis of the spine due to an osteochondrosis (Scheuermann's disease). This condition occurs typically in the thoracic spine, but is also seen at the thoracolumbar junction. The thoracolumbar form of Scheuermann's disease, which encompasses both the thoracic and lumbar regions of the spine, is more common in athletes than non-athletes and also is considered to be associated more frequently with back pain.¹⁶ Children can present with acute pain. It usually presents in later years as an excessive thoracic kyphosis in association with a compensatory excessive lumbar lordosis.

The typical radiographic appearance of Scheuermann's disease is shown in Figure 44.4. This demonstrates irregularity of the growth plates of the vertebrae. The radiological diagnosis of Scheuermann's disease is made on the presence of wedging of 5° or more at three adjacent vertebrae.

Management is aimed at preventing progression of the postural deformity and involves a combination of joint mobilisation, massage therapy to the thoracolumbar fascia, stretching of the hamstring muscles and abdominal

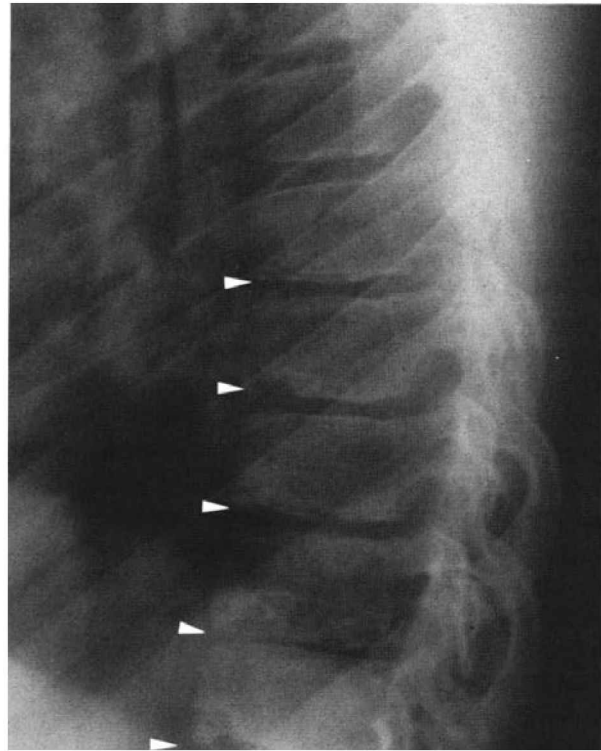


Figure 44.4 Radiographic appearance of Scheuermann's disease

muscle strengthening. A brace may be worn to decrease the thoracic kyphosis and lumbar lordosis. Surgery may be indicated if the kyphosis is greater than 50° or if signs of spinal cord irritation are present.

Hip pain

Hip pain is a more common presenting symptom in the younger athlete than the mature adult. The causes of hip pain in the younger athlete are shown in Figure 44.5. There are a number of possible causes of persistent hip pain and decreased range of motion in the younger athlete, in addition to those seen in the older athlete such as FAI, labral tears, stress fractures and snapping hip (all discussed in Chapters 31 and 32). MRI plays an important role in the diagnosis of the young athlete with hip pain.

Apophysitis

A number of large musculotendinous units attach around the hip joint. Excessive activity can result in a traction apophysitis at one of these sites, usually the ischial tuberosity at the attachment of the hamstring muscles, the anterior inferior iliac spine at the attachment of the rectus femoris, the anterior superior iliac spine at the attachment of the sartorius, or the iliopsoas attachment

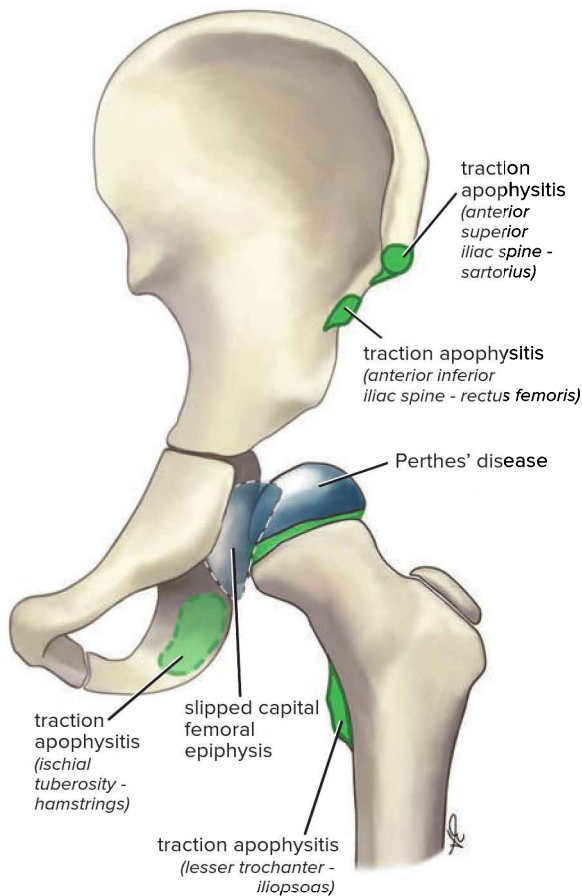


Figure 44.5 Causes of hip pain in children

to the lesser trochanter. Conservative therapy, including relative rest and modification of the athlete's activity level and strengthening, is usually effective.¹⁸

Perthes' disease

Perthes' disease is an osteochondritis affecting the femoral head. Perthes' disease is usually unilateral. It typically affects children between the ages of 4 and 10 years, is more common in males, and may be associated with delayed skeletal maturation. It presents as a limp or low-grade ache in the thigh, groin or knee.¹⁹ On examination there may be limited abduction and internal rotation of the hip. Radiographs vary with the stage of the disease, but may show increased density and flattening of the femoral capital epiphysis (Fig. 44.6).

Management consists of rest from aggravating activity and range of motion exercises, particularly to maintain abduction and internal rotation. The age of the child and the severity of the condition according to radiographic findings (Herring or Catarell scale) will affect the intensity of the management. For initial stages in children under 6 years of age, pain killers, and avoiding jumping and



Figure 44.6 Radiographic appearance of Perthes' disease

running are sufficient to control the disease. In older children (over 6 years) with lateral pillar collapse of the femoral head over 50%, surgical treatment is indicated.

Various surgical techniques are used to cover the femoral head and protect it from further damage. Most frequently, femoral osteotomy or iliac augmentation techniques (or both) are used. In borderline cases, usage of orthoses is indicated. It is important for parents to prevent their children from engaging in sports that involve running and jumping, especially during the healing period of the femoral head (lasting approximately 2–3 years).

The condition usually resolves and return to sport is possible when the child is symptom-free and radiographs show some improvement. The main long-term concern is the development of osteoarthritis due to irregularity of the joint surface.

Slipped capital femoral epiphysis

A slipped capital femoral epiphysis (SCFE) may occur in older children, particularly between 12 and 15 years,²⁰ but can also be observed in younger children. Radiographs in SCFE have a similar appearance to a Salter-Harris type I fracture.

A slipped capital femoral epiphysis is associated with endocrinological disorders, biomechanical stress on the proximal femoral physis and high body mass index (BMI). It occurs typically in overweight boys who tend to be late-maturing, and thus is of increased concern given the trend towards adolescent obesity and selection for heavy boys in sports like American football and rugby. It has also been suggested that increased physical stress associated with intense sports participation may precipitate this condition.²¹

The slip itself may occur suddenly or, more commonly, as a gradual process. There is sometimes associated pain, frequently in the knee, but the most common presenting symptom is a limp. Examination reveals shortening and external rotation of the affected leg. Hip abduction and internal rotation are reduced. During flexion the hip moves into abduction and external rotation.

Bilateral involvement is common (approximately 20% of cases). Slips may compromise the vascular supply to the femoral head and lead to avascular necrosis.



Figure 44.7 Radiographic appearance of slipped capital femoral epiphysis

PRACTICE PEARL

Radiographs show widening and irregularity of the growth plate, cystic changes in the proximal femoral metaphysis and a line continued from the superior surface of the neck of the femur does not intersect the growth plate (Fig. 44.7).

Avascular necrosis is the most common and severe complication of a slipped capital femoral epiphysis and requires orthopaedic assessment. A gradually progressing slip is an indication for surgery. The aim is to prevent further slipping of the epiphysis by transcervical fixation using one or more cannulated screws. An acute severe slip occurs occasionally. This is a surgical emergency.

Irritable hip

'Irritable hip' or transitory synovitis is common in children, but should be a diagnosis of exclusion. It is often related to a respiratory, often viral, infection. The child presents

with a limp and pain that may not be well localised. Examination reveals painful restriction of motion of the hip joint, particularly in extension and/or abduction in flexion. Radiographs and blood tests are usually normal. Ultrasound may demonstrate fluid in the joint capsule and MRI can help in challenging cases.

In the majority of cases, a specific cause is never identified and the pain settles after a period of bed rest and observation.²²

Knee pain

Knee pain, especially anterior knee pain, is a common presentation in the younger athlete. Common causes of anterior knee pain include: Osgood-Schlatter disease, Sinding-Larsen-Johansson disease, patellofemoral joint pain (Chapter 36), patellar tendinopathy (Chapter 36), and referred pain from the hip.

Osgood-Schlatter disease

Osgood-Schlatter disease is an osteochondritis that occurs at the growth plate of the tibial tuberosity. Repeated

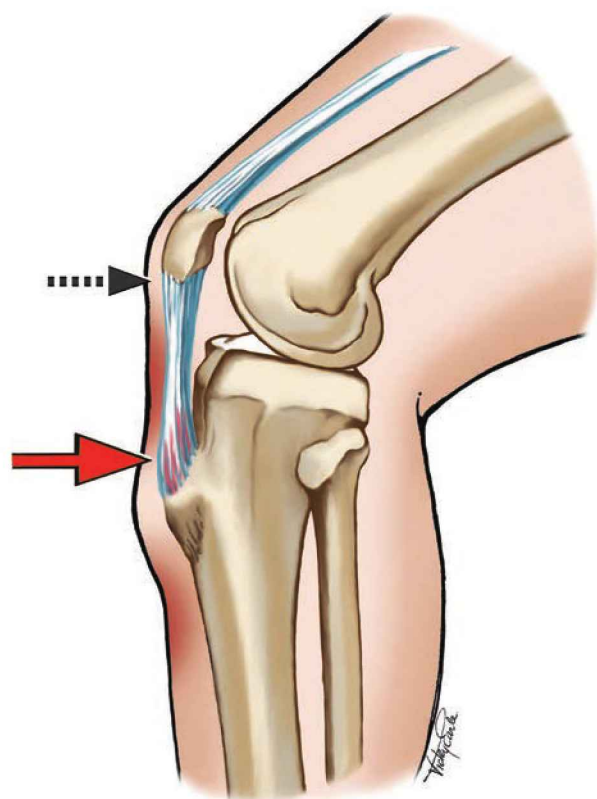


Figure 44.8 Sites of maximal tenderness in Sinding-Larsen-Johansson disease (black arrow) and Osgood-Schlatter disease (red arrow)

contraction of the quadriceps muscle mass may cause softening and partial avulsion of the developing secondary ossification centre.

This condition is extremely common in adolescents at the time of the growth spurt. It is usually associated with repeated forced knee extension, especially in sports involving running and jumping, such as basketball, football or gymnastics. Pain around the tibial tuberosity is aggravated by exercise.

Examination reveals tenderness over the tibial tuberosity (Fig. 44.8). There may be associated tightness of the surrounding muscles, especially the quadriceps. The presence of excessive subtalar pronation may predispose to the development of this condition.

The diagnosis of Osgood-Schlatter disease is clinical and radiographs are usually not required. In cases of severe anterior knee pain with more swelling than expected, a radiograph may be indicated to exclude bony tumour. Although all bone tumours are rare, the knee is a site of osteogenic sarcoma in the 10-30 year age group. The typical radiographic appearance of Osgood-Schlatter disease is shown in Figure 44.9.



Figure 44.9 Radiographic appearance of Osgood-Schlatter disease

Osgood-Schlatter disease is a self-limiting condition that settles at the time of bony fusion of the tibial tubercle. Its long-term sequel may be a thickening and prominence of the tubercle. Occasionally, a separate fragment develops at the site of the tibial tubercle. Athletes and parents need to understand the nature of the condition as symptoms may persist up to full maturity.

Management

Management of this condition requires activity modification. While there is no evidence that rest accelerates the healing process, a reduction in activity will reduce pain.

PRACTICE PEARL

As this condition occurs in young athletes with a high level of physical activity, it may be useful to suggest they eliminate one or two of the large number of sports they generally play. There is no need to rest completely. Pain should be the main guide as to the limitation of activity.

Symptomatic management via relative load reduction is key. The athlete may consider a stretching program and some massage therapy to the quadriceps muscle. Muscle strengthening can be introduced as pain allows.

Correction of any predisposing biomechanical abnormality, such as excessive subtalar pronation, is necessary. Neither injection of a corticosteroid agent nor surgery is required. Very occasionally, the skeletally mature person

may continue to have symptoms (e.g. with work-related kneeling) due to non-union. The separate fragment should then be excised.

Sinding-Larsen–Johansson disease

This is a similar condition to Osgood-Schlatter disease. It affects the inferior pole of the patella at the superior attachment of the patellar tendon (Fig. 44.8). It is much less common than Osgood-Schlatter disease, but the same principles of management apply.

Patellar tendinopathy

Although symptomatic tendinopathy was thought to be rare in children, Professor Jill Cook showed that patellar tendinopathy is prevalent in junior basketball players.^{23, 24} Management is as per the tendinopathy management principles in adults (Chapter 36).

Referred pain from the hip

Conditions affecting the hip, such as a slipped capital femoral epiphysis or Perthes' disease, commonly present as knee pain due to the innervation by the obturator nerve. Examination of the hip joint is mandatory in the assessment of any young athlete presenting with knee pain.

Popliteal cysts

Popliteal cysts (called 'Baker's cysts') are cystic formations filled with gelatinous mass developing in the popliteal fossa. The usual presenting complaint in children is the presence of a mass at the back of the knee. The most common site of origin is the bursa of the gastrocnemius and semimembranosus muscle. The cyst often appears gradually and may be fairly large when first noticed. Occasionally, it is found after an injury. Popliteal cysts in children are usually minimally symptomatic or asymptomatic, and not related to an intra-articular pathologic process.

Physical examination reveals a firm cystic mass in the popliteal fossa, often medially located and usually distal to the popliteal crease. It is most prominent when the knee is hyperextended with the patient in a prone position. Plain radiographs and ultrasound are the imaging techniques of choice. MRI should be obtained if an atypical history or abnormal physical findings are present.

In the vast majority of cases, popliteal cysts do not require treatment. The cyst usually resolves spontaneously over a period of months to a few years. Surgical excision of a popliteal cyst is indicated only when symptoms are severe and limit the range of motion of the knee, and have not resolved with at least several months of follow-up.

Less common causes of knee pain

Osteochondritis dissecans may affect the knee. This generally presents with intermittent pain and swelling

of gradual onset. Occasionally, it may present as an acute painful locked knee. This acute presentation is associated with haemarthrosis and loose body formation. Radiographs may reveal evidence of a defect at the lateral aspect of the medial femoral condyle. Osteochondritis dissecans requires orthopaedic referral for possible fixation of the loosened fragment or removal of the detached fragment.

In juvenile rheumatoid arthritis (Still's disease) of the knee, there is a persistent intermittent effusion with increased temperature and restricted range of motion. There may be a family history of rheumatoid arthritis. Investigation requires serological examination, including measuring the level of rheumatoid factor and the erythrocyte sedimentation rate and, if indicated, serological examination of joint aspirate. The child's activity should be adapted to avoid using the painful joints while exercising other body parts and promoting cardiovascular fitness.

A differential diagnosis in paediatric arthritis that is relatively rare but is increasing in the developed world is acute rheumatic fever.²⁵ As there may be no history of sore throat and carditis may be silent, the diagnosis can only be made if the practitioner maintains an index of suspicion for this condition. Investigations should include markers of inflammation (erythrocyte sedimentation rate, C-reactive protein), serology for streptococci (anti-streptolysin-O titre, anti-DNaseB titres) and echocardiography. Penicillin and aspirin taken orally remain the mainstays of management.

A partial discoid meniscus may cause persistent knee pain and swelling in the adolescent athlete. A complete discoid meniscus is characterised by a history of clunking in the younger child (approximately 4 years). There is usually marked joint line tenderness, and a palpable snapping at the joint line with flexion/extension and/or rotation. MRI will demonstrate the lesion. Arthroscopy may be diagnostic and therapeutic.

Adolescent tibia vara (Blount's disease) is an uncommon osteochondritis that affects the proximal tibial growth plate. It usually affects tall, obese children around the age of 8–10 years. It is generally unilateral and radiographs show a reduced height of the medial aspect of the proximal tibial growth plate. Surgery may be required to correct any resultant mechanical abnormality.

Painless abnormalities of gait

It is not uncommon for a child to present with an abnormality of gait. The child is typically brought in by a parent who has noticed an unusual appearance of the lower limb or an abnormal gait while walking or running. The child may complain of foot pain (see below). However, in many instances, the abnormal gait is painless.

It is not sufficient to say that the child will 'grow out of it'. The child requires a thorough biomechanical

Anterior cruciate ligament (ACL) injuries in the skeletally immature athlete

with HÅVARD MOKSNES

Instability and functional impairments following ACL tears in skeletally immature children have been increasingly recognised.^{26–32} Intrasubstance ACL ruptures are most worrisome due to the serious long-term health effects of potential early osteoarthritis.³³ Furthermore, the open growth plates on both sides of the knee joint warrant particular caution before surgical interventions with ACL reconstruction are performed.^{34, 35}

Risk factors

It seems that boys may be more prone to rupturing their ACL before maturity, while girls have an increased risk through and after puberty,^{36, 37} and most authors state that the incidence of these injuries is rising. However, no epidemiological studies are available with historical or new data to support this assumption. The perceived increased occurrence of ACL tears in children and adolescents may be due to higher participation and early specialisation in sports; however, it can also be caused by increased awareness and advances in diagnostic methods.

Treatment

Treatment algorithms for ACL ruptures in skeletally immature children vary around the world.^{28–40}

PRACTICE PEARL

One of three different treatment algorithms is traditionally recommended to skeletally immature children after ACL injury:^{38, 39, 41}

- a transphyseal (adult) surgical reconstruction
- non-operative treatment with active rehabilitation and an optional delayed ACL reconstruction if specific criteria are met
- a physeal-sparing ACL reconstruction.

Unfortunately, the methodological quality in research on treatment of ACL injuries in younger populations has been poor.⁴² Thus, there is a need for adequately sized prospective multicentre studies to heighten the scientific base for which treatments should be offered to children after suffering an ACL injury. Specific decision criteria with regard to which of these algorithms a child should be recommended have not been established, and treatment decisions are traditionally based on the experiences and practice of the individual orthopaedic surgeon or institution.

(i) Surgical reconstruction

Surgical reconstruction of the ACL in a skeletally immature patient is usually advocated to provide ligamentous knee joint stability, and to potentially protect the menisci from subsequent injury. However, surgical treatment may also damage the epiphyseal growth plates and result in various growth disturbances.^{34, 35, 43} Furthermore, development of the graft during the remaining skeletal growth is unknown, and different authors have discussed the possibility of increased risk of re-injuries in adulthood due to impaired biomechanical properties.^{31, 44, 45}

(ii) Primary active rehabilitation without surgical reconstruction

Conversely, primary active rehabilitation without surgical reconstruction has been documented to give favourable functional outcomes for a majority of children who have undergone supervised active rehabilitation programs.³⁰ Still, between one-third and half of these children report a change in their preferred sports activities from level 1 to level 2 activities (e.g. abandoning pivoting sports). Studies reporting comparable numbers following surgical treatment have not yet been reported. Either way, paediatric ACL injuries should be thoroughly assessed on an individual basis and treatment algorithms should be presented in a balanced fashion to the child and his or her parents.

(iii) Physeal-sparing ACL reconstruction

Intercondylar tibial eminence avulsion fractures involving the ACL are frequent and have long been assumed to be the most common injury involving the ACL in skeletally immature children.⁴⁶ Since a paper on surgical treatment of tibial spine avulsion injuries with involvement of the ACL from Roth in 1928,⁴⁷ consensus has developed that the majority of these injuries should be treated surgically with fixation to the bone without intervening with the epiphyseal growth plate.^{48, 49}

Surgery or rehabilitation?

The clinical challenges with treatment for paediatric ACL injuries are related to the weighing of risks and benefits between primary surgical treatment and primary active rehabilitation without surgical intervention.⁵⁰ Regardless of whether the child is advised to undergo surgical treatment or not, there is substantial support in the literature that supervised rehabilitation should be performed exhaustively through all phases before the ultimate treatment decision is taken (Chapter 35).

Rehabilitation

Paediatric rehabilitation has to be performed in close collaboration with the parents. Exercises and goals have to be adjusted compared to traditional rehabilitation

continued

protocols because children are not small adults, and they cannot be expected to perform unsupervised training independently. Rehabilitation exercises are primarily focused on neuromuscular stimulation and control, with less focus on muscular strength and hypertrophy.^{50, 51} Throughout the first two phases the child should be guarded from pivoting activities, and possibly should also wear a protective brace in school and training. Training sessions including exercises to facilitate proper alignment and adequate landing techniques have been successfully implemented in injury prevention

programs,^{52–54} and are also recommended through phases 2 and 3 of an ACL rehabilitation protocol.

Similarly, rehabilitation after surgical interventions has to be thorough and individualised with regard to the child's physiological and psychological maturity to achieve successful outcomes. Additionally, a longer time from surgery to return to sport should be advocated as emerging data from international registries suggest that young athletes have a high risk of suffering a second ACL injury following an ACL reconstruction.^{55–59}

assessment, which may reveal a structural abnormality. The most common biomechanical problems in children are rotational abnormalities originating from the hip and the tibia causing either in-toed or out-toed gait.

If the child is asymptomatic and biomechanical abnormalities are not marked, no treatment is indicated. If abnormalities are marked or if the child is symptomatic, management may involve physiotherapy, as well as the use of braces or night splints when the child is very young and during periods of rapid growth. In the older child, orthoses can be used to compensate for the deformity.

Foot pain

Foot pain of gradual onset is a common presenting symptom in the younger athlete. The causes of foot pain are either acquired and congenital foot deformities such as flat feet, metatarsus adductus, clubfoot, vertical talus, pes calcaneovalgus or toe abnormalities, or osteochondroses of the foot such as Köhler's disease and Freiberg's disease. Examination of younger athletes with foot pain requires precise determination of the site of maximal tenderness.

Sever's disease

Heel pain is a common complaint in the skeletally immature athlete. Sever originally described⁶⁰ this condition as an inflammatory condition to the apophysis (Sever's apophysitis). It is a self-limited disease.

The patient complains of activity-related pain and examination reveals localised tenderness and swelling at the site of insertion of the Achilles tendon. There may be tightness of the gastrocnemius or soleus muscles, and dorsiflexion at the ankle is limited. Biomechanical examination is necessary. Radiographic examination is usually not required except in persistent cases.

Management consists of activity modification so that the child becomes pain-free. The patient should be advised that the condition will typically always settle, usually within 6–12 months, but occasionally symptoms will persist for as long as 2 years. A heel raise should be inserted in shoes. Stretching of the calf muscles is also advisable.

Any biomechanical abnormalities should be corrected. Orthoses may be required. Strengthening exercises for the ankle plantarflexors should be commenced when pain-free and progressed as symptoms permit. Corticosteroid injections and surgery are contraindicated in this condition.

There is MRI evidence that this condition is not an inflammation within the apophysis, but rather is a chronic (repetitive) injury to the actively remodelling trabecular metaphyseal bone.^{61, 62} This condition should be treated as a stress fracture with a limited period of partial or non weight-bearing⁶¹ and temporary discontinuation of the offending activity.

Tarsal coalitions

Congenital fusions of the bones of the foot may be undetected until the child begins participation in sports. The most common form is a bony or cartilaginous bar between the calcaneus and navicular bone. The second most common coalition is between the calcaneus and talus. Calcaneocuboid coalition is the least common form. There is often a family history. The adolescent may present with midfoot pain, after recurrent ankle sprains or after repetitive running and jumping. The pain may be associated with a limp.

Examination reveals restriction of subtalar joint motion. There may be a rigid flat-foot deformity. Radiographs taken at 45° oblique to the foot may confirm the diagnosis, but if these are normal and clinical suspicion persists, a CT scan or MRI should be obtained.

Management may require orthotic therapy. Surgical excision may be necessary in a young patient with severe symptoms or after failure of conservative therapy. The bony or cartilaginous bar may recur after surgery.

Köhler's disease

Köhler's disease is a form of osteochondrosis affecting the navicular bone seen in young children, especially between the ages of 2 and 8 years, and more frequently in boys. The exact cause of Köhler's disease is not known. It has

been suggested in the literature that because of its late ossification relative to the other tarsal bones, the navicular is vulnerable to mechanical compression injury.

The child complains of pain over the medial aspect of the navicular bone and often develops a painful limp. Tenderness is localised to the medial aspect of the navicular bone. Radiographs reveal typical changes of increased density and narrowing of the navicular bone. Management in a walking cast for 6 weeks may accelerate relief of the symptoms. Orthoses should be used if biomechanical abnormalities are present. The natural history of the disease is spontaneous resolution of clinical symptoms and the radiographic abnormalities over time.

Apophysitis of the tarsal navicular bone

Pain on the medial aspect of the tarsal navicular in the older child may result from a traction apophysitis at the insertion of the tibialis posterior tendon to the navicular. This condition is often associated with the presence of an accessory navicular (Fig. 43.16) or a prominent navicular tuberosity. Management involves modification of activity, local electrotherapy and NSAIDs, with orthoses to control excessive pronation if this is present.

The accessory navicular which is found in 2.5% of children can cause pain by itself. Throughout early childhood, this condition is not noticed. However, in adolescence, when the accessory navicular begins to calcify, the bump on the inner aspect of the arch becomes noticed. For most, it is never symptomatic. However, for some, there is some type of injury, whether a twist, stumble or fall, that makes the accessory navicular symptomatic.

Apophysitis of the fifth metatarsal

A traction apophysitis at the insertion of the peroneus brevis tendon to the base of the fifth metatarsal is occasionally seen. Examination reveals local tenderness and pain on resisted eversion of the foot. Management consists of modification of activity, stretching and progressive strengthening of the peroneal muscles.

Freiberg's disease

Freiberg's disease (infarction) is an osteochondrosis causing collapse of the articular surface and adjacent bone of the metatarsal head. It is commonly thought to be due to avascular necrosis of the metatarsal head as demonstrated by destructive radiographic changes. The second metatarsal is most commonly involved (especially in ballet dancers), the third occasionally, and the fourth rarely. It occurs most frequently in adolescents over the age of 12 years.

Standing on the forefoot aggravates pain. The head of the second metatarsal is tender and there is swelling around the second metatarsal joint. Radiographs reveal a flattened head of the metatarsal with fragmentation of the growth plate. However, these changes may lag well behind the symptoms. Isotopic bone scan or MRI are more sensitive investigations.

If Freiberg's disease is diagnosed early, management with activity modification, padding under the second metatarsal and footwear modification to reduce the pressure over the metatarsal heads may prove successful. If the symptoms persist, surgical intervention may be necessary.

Flat foot

The exact incidence of flat foot in children is unknown. It is undoubtedly one of the most common 'deformities' evaluated by sports medicine clinicians. Most often, flat foot is a reflection of ligamentous laxity in the foot resulting in an abnormally low or absent arch.

In asymptomatic paediatric patients with a hypermobile flat foot, no treatment is indicated. Orthopaedic shoes, including various heel modifications, moulded heel cups, medial arch support and other orthoses, have traditionally been prescribed, even though there is no scientific evidence that such measures are efficacious.

In symptomatic patients, arch support and other orthoses may be of benefit. In some cases, where bony abnormalities caused by tarsal coalition (rigid type of flat foot) are present, surgical intervention is indicated if conservative treatment does not achieve satisfactory results.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

